

EXPT NO:3	CLEANSE DATA BY HANDLING MISSING VALUES, OUTLIERS, AND INCONSISTENCIES USING TABLEAU AND POWERBI
DATE: 10.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What are the potential impacts of missing values in a dataset, and what are some common strategies to address them in Tableau and Power BI?

Missing values can severely distort statistical analysis by skewing averages and leading to inaccurate business conclusions. They also reduce the overall reliability of predictive models and visualizations by creating incomplete data narratives. In Power BI, these gaps are typically addressed using the Power Query Editor to filter out nulls or replace them with conditional baseline values. Tableau handles these issues effectively by allowing users to exclude null indicators directly from visualizations or by utilizing the ZN() function to convert empty numerical fields to zero.

2. What is the importance of identifying and addressing outliers in a dataset? How can outliers affect data analysis and model performance?

Identifying outliers is crucial because they represent abnormal data points that dramatically skew statistical averages and variance calculations. When left unaddressed, these extreme values can heavily bias predictive models, causing them to perform poorly on standard data entries. Addressing them ensures that the resulting analysis accurately reflects the true underlying trends rather than being hijacked by isolated, extreme anomalies. Analysts typically manage these outliers by either capping their maximum values or removing them entirely to maintain the structural integrity of the analysis.

3. What types of data inconsistencies might you encounter during data cleansing, and how can Tableau and Power BI help in resolving them?

Data inconsistencies frequently manifest as mixed formatting, mismatched text cases, duplicate records, or incorrectly assigned data types. These errors prevent accurate aggregation and disrupt the ability to merge disparate tables into a cohesive relational model. Power BI resolves these issues effortlessly through Power Query features like text capitalization, column splitting, and automated duplicate removal tools. Tableau provides robust capabilities through built-in string functions and grouping tools, allowing users to standardize formats and consolidate similar textual values logically.

4. How do you identify patterns in data using visualizations to spot missing values, outliers, and inconsistencies?

Visualizations provide an immediate, intuitive method for spotting data irregularities that might remain completely hidden within raw tabular rows. Scatter plots are highly effective for identifying outliers, as extreme data points will visually separate themselves far away from the primary data clusters. Bar charts and tables quickly expose missing values by generating blank categories or displaying null indicator labels directly along the axes. Time-series line charts also clearly highlight inconsistencies by displaying sudden, unnatural spikes or severe drops that signal formatting errors.

5. What is the role of calculated fields in data cleansing, and how can they be used to handle missing values and outliers in Tableau/Power BI?

Calculated fields act as customizable data transformation tools, enabling analysts to write specific logic that cleanses information without altering the original source file. They are frequently used to handle missing values by utilizing conditional IF statements that automatically replace nulls with a designated baseline metric or average. For outliers, calculated fields can establish mathematical upper and lower thresholds, clamping extreme values to a maximum limit to prevent them from skewing the visual scale. Both platforms rely heavily on these customized formulas to create dynamic, logically sound dimensions for accurate reporting.

IN-LAB EXERCISE:

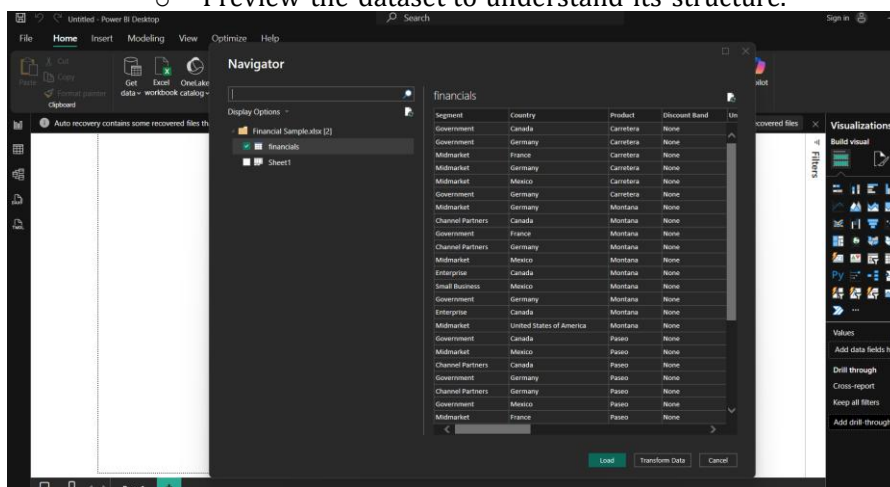
OBJECTIVE:

To clean a given dataset (e.g., Superstore Sample in Tableau and Financials/Sales Data in Power BI) by identifying and resolving missing values, outliers, and inconsistencies using built-in tools, ensuring the dataset is accurate and ready for analysis.

PROCEDURE (In PowerBI):

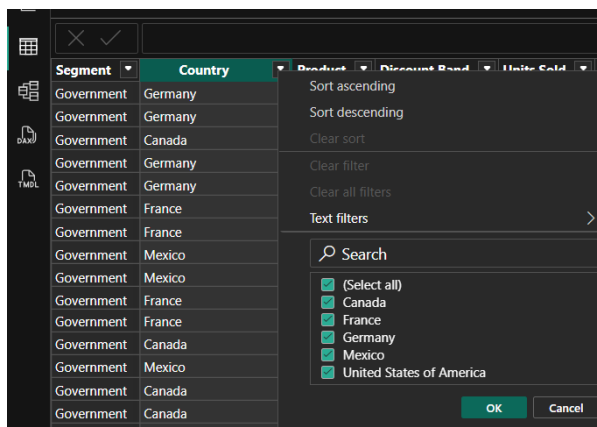
1. Open the Tool and Connect Dataset

- Open Power BI Desktop.
- Load the built-in dataset (Financials in Power BI).
- Preview the dataset to understand its structure.



2. Initial Inspection

- Explore fields such as Sales, Profit, Region, or Product Category.
- Check for nulls, inconsistent text values (e.g., "South Region" vs. "South"), or date formatting issues.

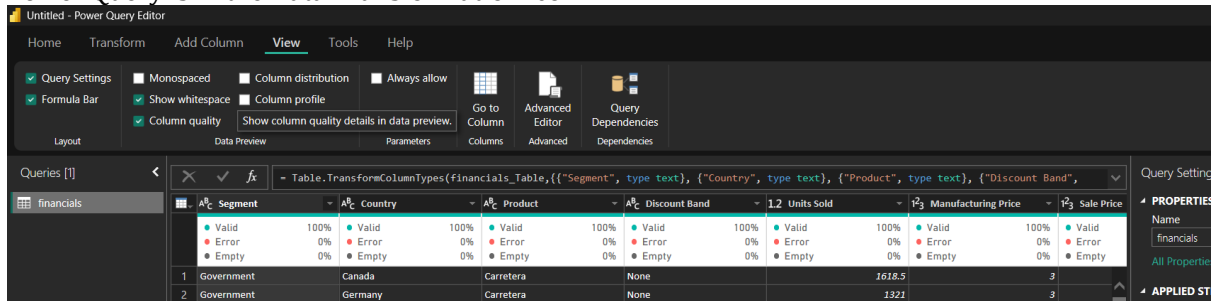


Inspected the Region or Country attribute where "United States of America" need to be standardized to ensures data consistency, prevents aggregation errors, and saves valuable space on your dashboard's charts and visuals.

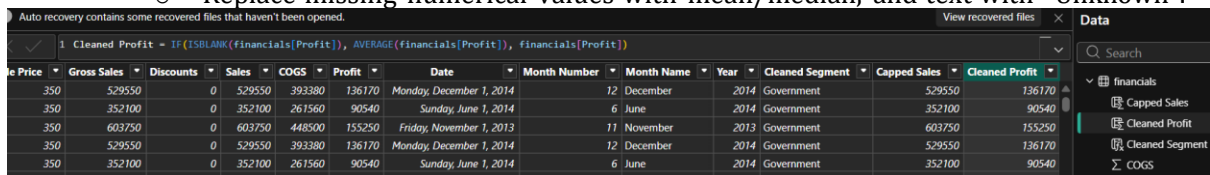
3. Handling Missing Values

- Use Power Query (Power BI) to highlight nulls.

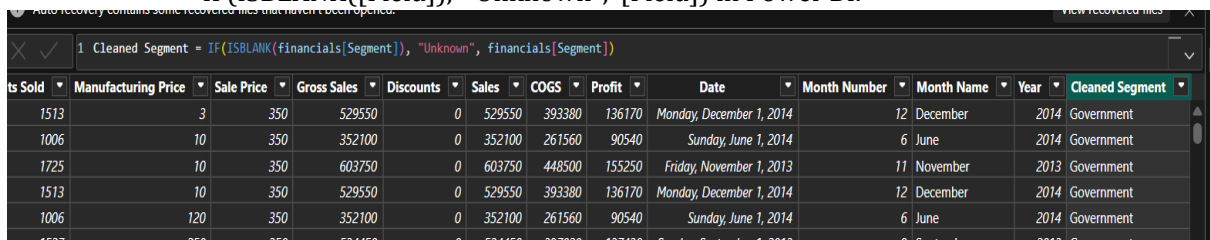
Power Query is in the Data Transformation icon.



- Replace missing numerical values with mean/median, and text with "Unknown".

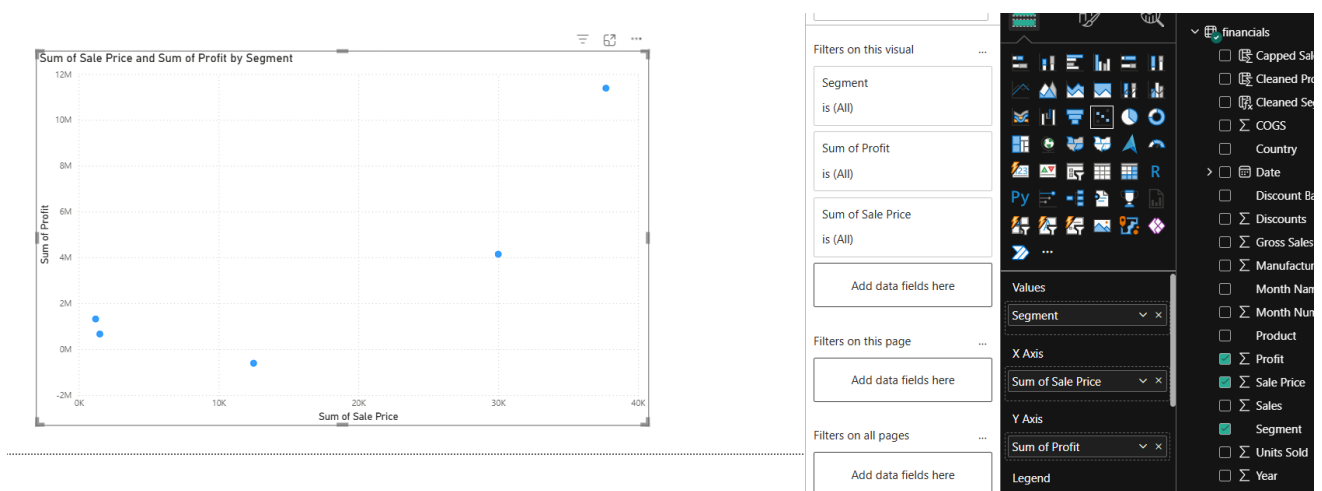


- Use calculated fields: IFNULL([Field], Default) in Tableau or IF(ISBLANK([Field]), "Unknown", [Field]) in Power BI.



4. Detecting and Managing Outliers

Create scatter or box plots for Sales and Profit to identify extreme values.



Apply cap using Calculated columns

uto recovery contains some recovered files that haven't been opened.

View recovered files X Data

1 Capped Sales = IF(financials[Sales] > 1000000, 1000000, financials[Sales])

Sales	Discounts	Sales	COGS	Profit	Date	Month Number	Month Name	Year	Cleaned Segment	Cleaned Profit	Column	Capped Sales
529550	0	529550	393380	136170	Monday, December 1, 2014	12	December	2014	Government	136170		529550
352100	0	352100	261560	90540	Sunday, June 1, 2014	6	June	2014	Government	90540		352100
603750	0	603750	448500	155250	Friday, November 1, 2013	11	November	2013	Government	155250		603750
529550	0	529550	393380	136170	Monday, December 1, 2014	12	December	2014	Government	136170		529550
352100	0	352100	261560	90540	Sunday, June 1, 2014	6	June	2014	Government	90540		352100

financials

- Capped Sales
- Cleaned Profit
- Cleaned Segment
- COGS

Flag them with labels for review.

1 Outlier Flag = IF(financials[Sales] > 1000000, "Review", "Normal")

Units	Sales	COGS	Profit	Date	Month Number	Month Name	Year	Cleaned Segment	Cleaned Profit	Column	Capped Sales	Outlier Flag
0	529550	393380	136170	Monday, December 1, 2014	12	December	2014	Government	136170		529550	Normal
0	352100	261560	90540	Sunday, June 1, 2014	6	June	2014	Government	90540		352100	Normal
0	603750	448500	155250	Friday, November 1, 2013	11	November	2013	Government	155250		603750	Normal
0	529550	393380	136170	Monday, December 1, 2014	12	December	2014	Government	136170		529550	Normal
0	352100	261560	90540	Sunday, June 1, 2014	6	June	2014	Government	90540		352100	Normal
0	534450	397020	137430	Sunday, September 1, 2013	9	September	2013	Government	137430		534450	Normal
0	962500	715000	247500	Saturday, February 1, 2014	2	February	2014	Government	962500		962500	Normal
4235	419265	314600	104665	Saturday, March 1, 2014	3	March	2014	Government	104665		419265	Normal
4889.5	484060.5	363220	120840.5	Wednesday, October 1, 2014	10	October	2014	Government	120840.5		484060.5	Normal
7542.5	746707.5	560300	186407.5	Monday, December 1, 2014	12	December	2014	Government	186407.5		746707.5	Normal
7542.5	746707.5	560300	186407.5	Monday, December 1, 2014	12	December	2014	Government	186407.5		746707.5	Normal
3302.25	326922.75	245310	81612.75	Tuesday, April 1, 2014	4	April	2014	Government	81612.75		326922.75	Normal
4889.5	484060.5	363220	120840.5	Wednesday, October 1, 2014	10	October	2014	Government	120840.5		484060.5	Normal
19964	978236	741520	236716	Monday, December 1, 2014	12	December	2014	Government	236716		978236	Normal
19964	978236	741520	236716	Monday, December 1, 2014	12	December	2014	Government	236716		978236	Normal
20762	1017338	771160	246178	Tuesday, October 1, 2013	10	October	2013	Government	246178		1000000	Review
20139	986811	748020	238791	Wednesday, October 1, 2014	10	October	2014	Government	238791		986811	Normal

financials

- Capped Sales
- Cleaned Profit
- Cleaned Segment
- COGS
- Column
- Country
- Date
 - Discount Band
- Discounts
- Gross Sales
- Manufacturing Price
- Month Name
- Month Number
- Outlier Flag

5. Resolving Inconsistencies

- Use Group/Alias (Tableau) / Replace Values (Power BI) to standardize text.

A ^B C Country	A ^B C Product	A ^B C Discount Band	1.2 Units Sold
Valid	Valid	Valid	Valid
Canada			
Germany			
France			
Germany			
Mexico			
Germany			
Germany			
Canada			
France			
Germany			
Mexico			
Canada			

Replace Values

Replace one value with another in the selected columns.

Value To Find

United States of America

Replace With

USA

Advanced options

OK

- Normalize data formats, such as date fields or currency units.

Formatting Properties

Formatting: \$ % , .00 4

Properties: Data category: Uncategorized, Sort by column: Sort, Data groups: Groups, Manage relationships: Relationships

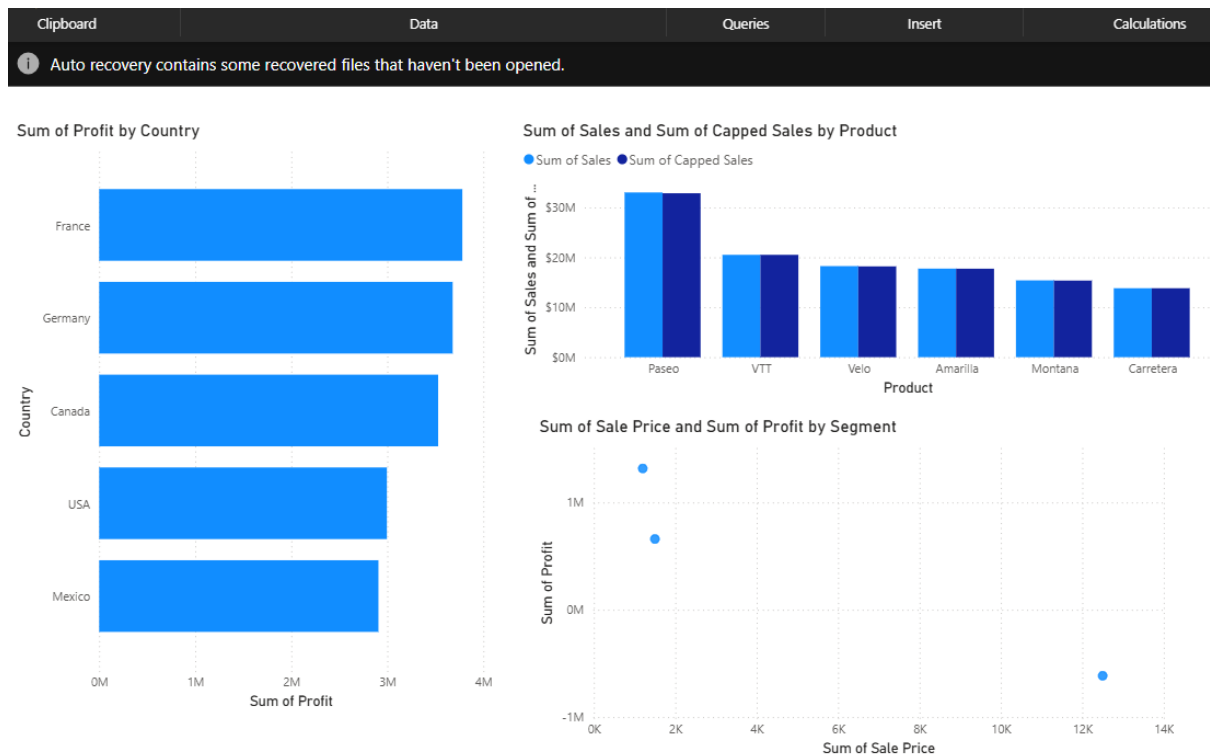
uto recovery contains some recovered files that haven't been opened.

Discount Band	Units Sold	Manufacturing Price	Sale Price	Gross Sales	Discounts	Sales
None	1513	3	350	529550	0	\$529,550.00
None	1006	10	350	352100	0	\$352,100.00
None	1725	10	350	603750	0	\$603,750.00
None	1513	10	350	529550	0	\$529,550.00
None	1006	120	350	352100	0	\$352,100.00
None	1527	250	350	534450	0	\$534,450.00

Formatting sales column to 2 decimal places along with \$ sign.

6. Validation with Visuals

- Build dashboards with histograms, bar charts, and box plots to visually confirm improvements.
- Compare before/after cleaning stages for accuracy.



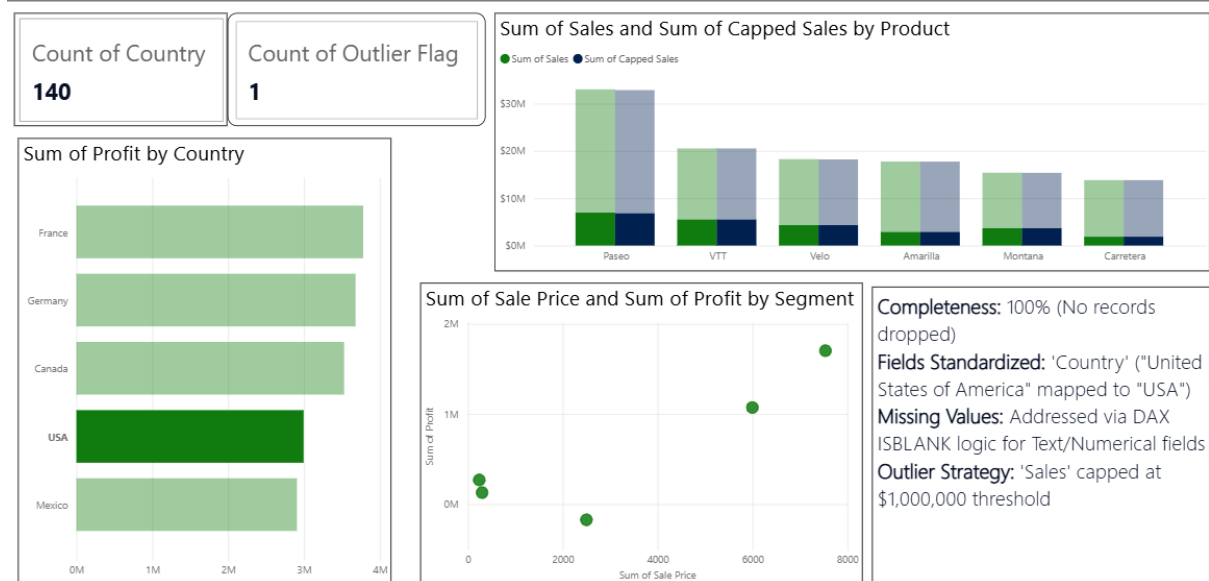
7. Save the Cleaned Dataset

- Save as .twbx (Tableau) / .pbix (Power BI).
- Optionally export cleaned data to .csv.

8. Summary Dashboard

- Create a visual summary of the cleaning process, highlighting:
 - Fields corrected.
 - Nulls/outliers removed or adjusted.
 - Clean dataset metrics (record count, completeness %).

Data Cleaning Summary Report



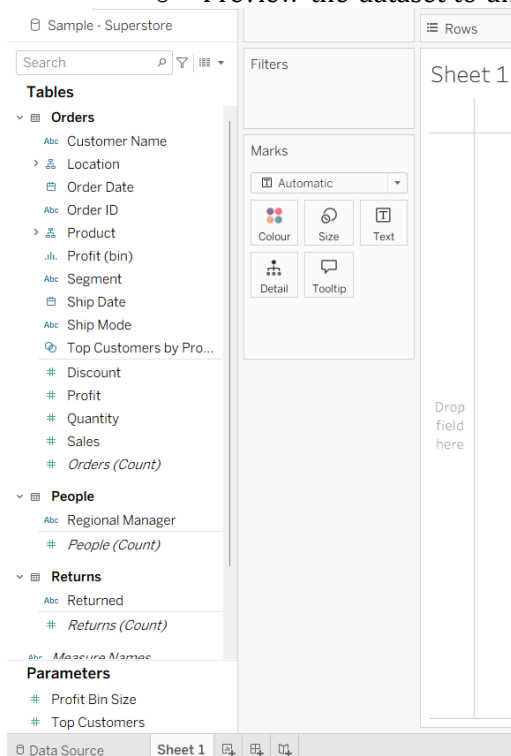
Github link on PowerBI dashboard: https://github.com/RohithJohnM/PowerBI-Dashboards/blob/main/Financial_Data_cleaning_in_powerBI.pbix

PDF link:

PROCEDURE (In Tableau):

1. Open the Tool and Connect Dataset

- Open Tableau Desktop
- Load the built-in dataset (Sample - Superstore in Tableau / Financials in Power BI).
- Preview the dataset to understand its structure.



2. Initial Inspection

- Explore fields such as Sales, Profit, Region, or Product Category.

The screenshot shows the Tableau interface with the 'Describe Field' dialog open for the 'Region' field. The dialog displays the following information:

- Role:** Discrete Dimension
- Type:** Database column
- Remote column:** [Orders].[Region]
- Remote type:** Unicode character string
- Contains NULL:** Unknown
- Locale:** United States(English)
- Sort flags:** Case-insensitive
- Column width:** Unknown
- Status:** Valid

The dialog also shows the domain of the field, which has 4 members: Central, East, South, and West. The 'Copy' button is highlighted.

- Check for nulls, inconsistent text values (e.g., "South Region" vs. "South"), or date formatting issues.

The screenshot shows the Tableau interface with a pivot table of Sales by Year and Quarter. The table is titled 'Sheet 1' and has the following data:

Year of O..	Quarter ..	
2023	Q1	75,972
	Q2	89,285
	Q3	145,535
	Q4	183,249
2024	Q1	70,392
	Q2	89,239
	Q3	130,449
	Q4	182,912
2025	Q1	94,840
	Q2	136,809
	Q3	145,539
	Q4	236,745
2026	Q1	125,289
	Q2	134,991
	Q3	198,184
	Q4	287,104

There is no null values the dataset has complete data integrity with no missing rows or unmapped records.

a. Handling Missing Values

- Use Data Interpreter (Tableau) / Power Query (Power BI) to highlight nulls.

calculated field to handle missing text value:

Filters

Marks

Automatic

Colour

Size

Text

Detail

Tooltip

SUM(Sales)

Rows

YEAR(Order Date)

QUARTER(Order ..

Sheet 1

Year of O..	Quarter ..	
2023	Q1	75,972
	Q2	89,285
	Q3	145,535
	Q4	183,249
2024	Q1	70,392
	Q2	89,239
	Q3	130,449
	Q4	182,912
2025	Q1	94,840
	Q2	136,809
	Q3	145,539
	Q4	236,745
2026	Q1	125,289
	Q2	134,991
	Q3	198,184
	Q4	287,104

Cleaned Segment

IFNULL([Segment], "Unknown")

The calculation is valid.

Apply

OK

calculated field to handle missing numerical values:

2024	Q4	183,249
	Q1	70,392
	Q2	89,239
	Q3	130,449
2025	Q4	182,912
	Q1	94,840
	Q2	136,809
	Q3	145,539
2026	Q4	236,745
	Q1	125,289
	Q2	134,991
	Q3	198,184
	Q4	287,104

Cleaned Sales

IFNULL([Sales], 0)

The calculation is valid.

Apply

OK

- Replace missing numerical values with mean/median, and text with "Unknown".

89,285
45,535
83,249
70,392
89,239
30,449
82,912
94,840
36,809
45,539
36,745
25,289
34,991
98,184
87,104

Cleaned Profit (Mean)

IFNULL([Profit], {AVG([Profit])})

The calculation is valid.

Apply

OK

- Use calculated fields: IFNULL([Field], Default) in Tableau or IF(ISBLANK([Field]), "Unknown", [Field]) in Power BI.

Replace Missing Text Values with "Unknown":

Year of O..	Quarter ..	
2023	Q1	75,972
	Q2	89,285
	Q3	145,535
	Q4	183,249
2024	Q1	70,392
	Q2	89,239
	Q3	130,449
	Q4	182,912
2025	Q1	94,840
	Q2	136,809
	Q3	145,539
	Q4	236,745
2026	Q1	125,289
	Q2	134,991
	Q3	198,184
	Q4	287,104

Cleaned Customer Name

IFNULL([Customer Name], "Unknown")

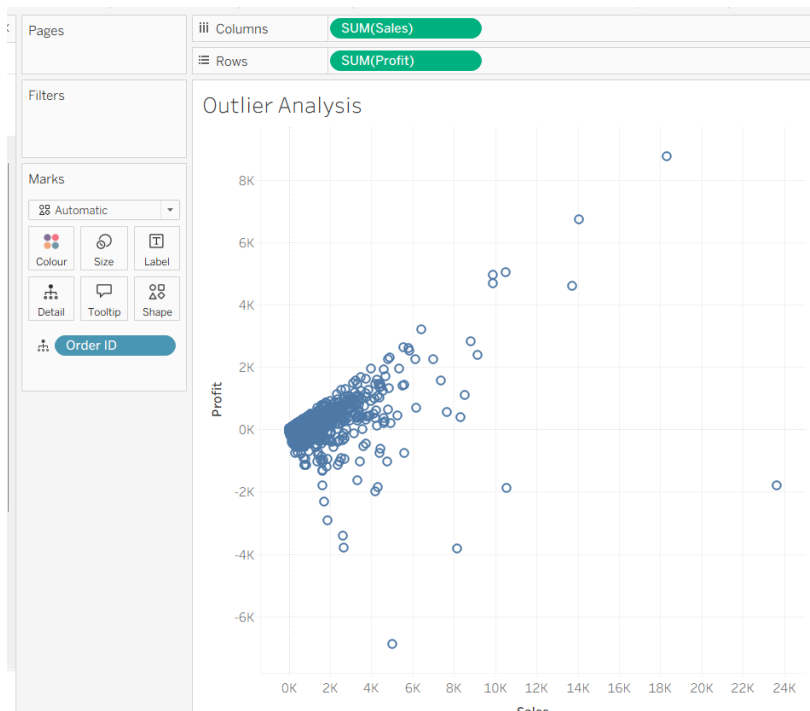
The calculation is valid.

Apply

OK

3. Detecting and Managing Outliers

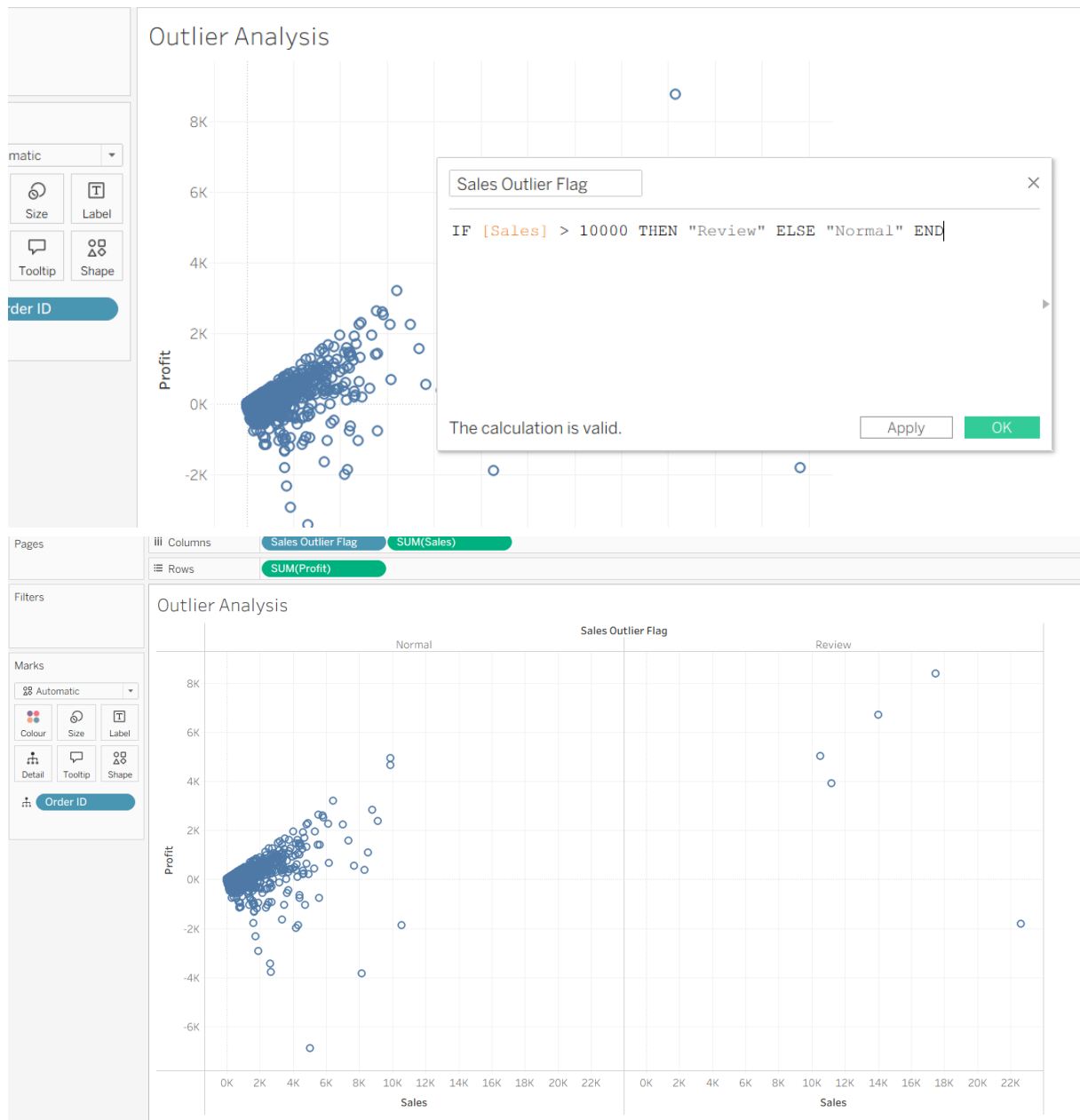
- Create scatter or box plots for Sales and Profit to identify extreme values.



- Choose strategies:
 - Filter them temporarily.
 - Apply caps using calculated columns.

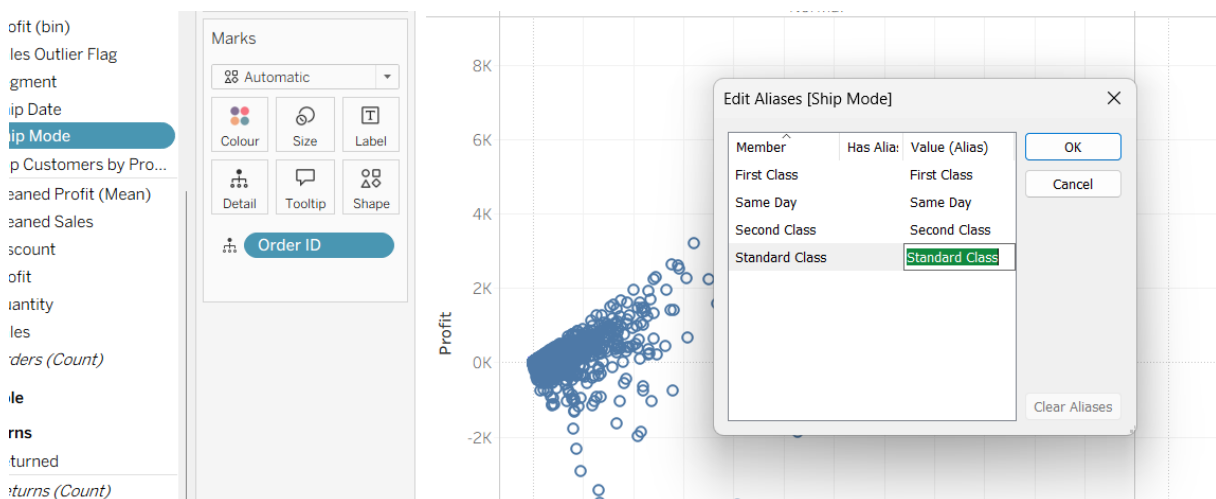
- Flag them with labels for review.

Capping strategy:



4. Resolving Inconsistencies

- Use Group/Alias (Tableau) / Replace Values (Power BI) to standardize text.

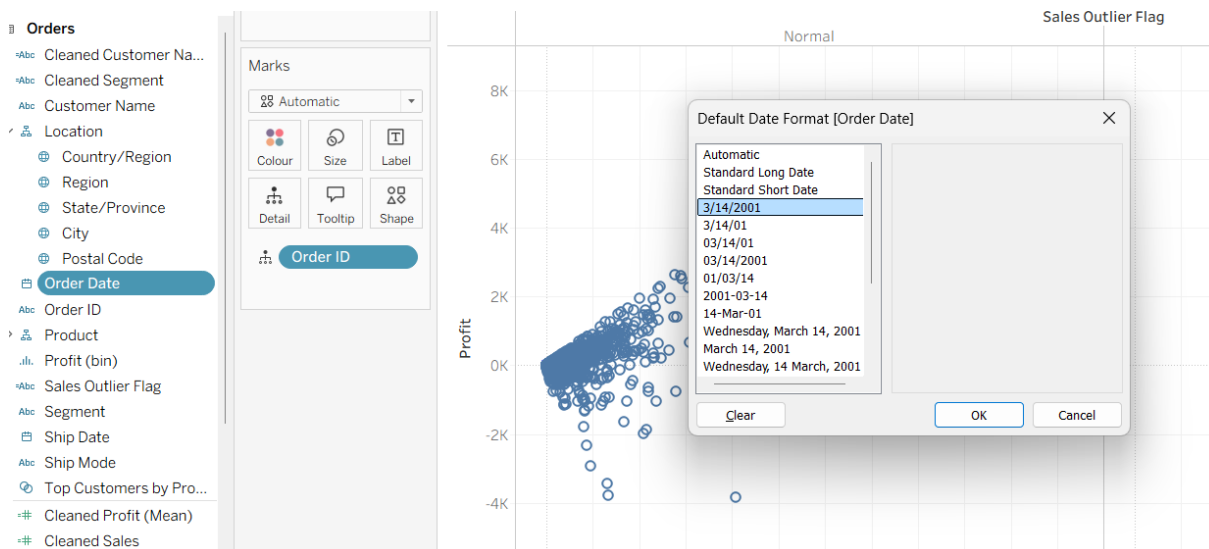


Edit Aliases [Ship Mode]

Member	Has Alia:	Value (Alias)
First Class		First Class
Same Day		Same Day
Second Class		Second Class
Standard Class	*	Universal Class

OK Cancel

- Normalize data formats, such as date fields or currency units.



5. Validation with Visuals

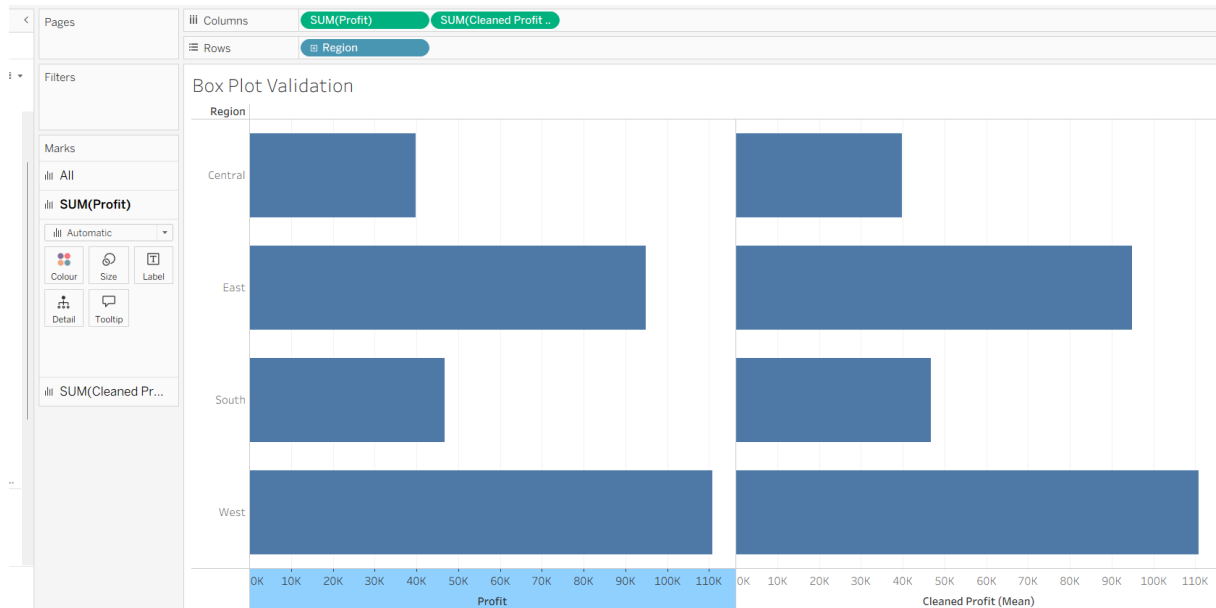
- Build dashboards with bar charts, and box plots to visually confirm improvements.

There is no change in Sales and Cleaned Sales:



The original Sales and Cleaned Sales values match identically because the dataset contains no missing sales values, confirming that the cleaning formula successfully preserved all valid data without corruption.

Same for the Profit attribute:



- Compare before/after cleaning stages for accuracy.

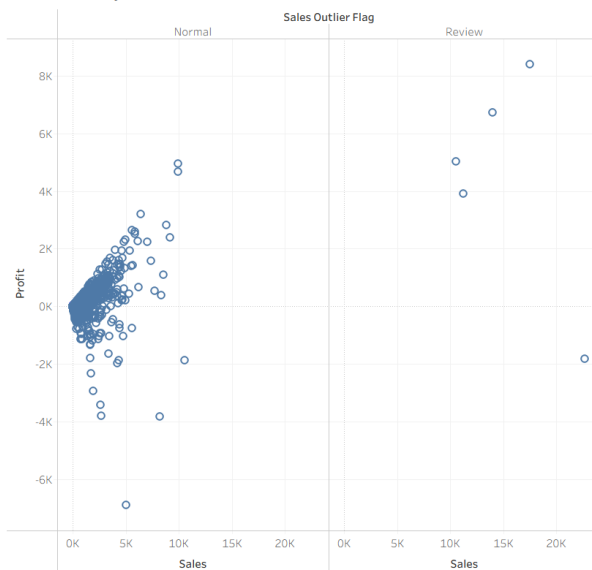
6. Save the Cleaned Dataset

- Save as .twbx (Tableau) / .pbix (Power BI).
- Optionally export cleaned data to .csv.

7. Summary Dashboard

- Create a visual summary of the cleaning process, highlighting:
 - Fields corrected.
 - Nulls/outliers removed or adjusted.
 - Clean dataset metrics (record count, completeness %).

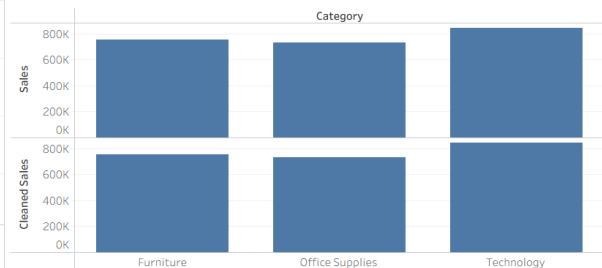
Outlier Analysis



Box Plot Validation



Bar Chart Comparison



Github link: https://github.com/RohithJohnM/Tableau-Dashboards/blob/main/SuperStore_Cleaning.twbx

Experiment Dataset:

1. Tableau: *Sample - Superstore*

2. Power BI: *Financials* or *Sales Data*

How to Access:

Tableau:

- **Navigate to:** *Start Page → Sample Workbooks*
- Or: *Help → Sample Data → Open Sample Data Folder*

Power BI:

- **Navigate to:** *Home → Get Data → Excel → Choose sample datasets (e.g., Financial Sample.xlsx)*
- Microsoft also hosts Power BI sample datasets online.

Fields Used:

1. Sales
2. Profit
3. Region
4. Category

5. Order Date / Transaction Date

POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1.What method did you use to replace missing values in numeric fields?

Missing numerical values in the dataset were addressed by implementing calculated fields that utilized conditional replacement logic. Specifically, functions such as IFNULL() were applied to substitute null values in measure fields with designated statistical baselines like the mean or zero. This approach ensured that incomplete entries were systematically populated without disrupting downstream analytical computations. Maintaining structured data integrity through this automated substitution prevented errors in visual aggregations. Consequently, the resulting dataset maintained complete analytical continuity across all numerical dimensions.

2.How were outliers handled in your dataset and what visual confirmed their correction?

Extreme values and outliers were identified by plotting measures such as sales against profit on a scatter plot and subsequently categorized using conditional threshold flags. Data points exceeding standard operational parameters were isolated and labeled as items requiring review or capping. An outlier analysis scatter plot visually confirmed these extreme values by mapping individual order IDs outside the primary data cluster. This visual verification allowed analysts to distinguish between normal density distributions and anomalous high-value transactions. Ultimately, the visualization provided clear evidence of how outliers were isolated for targeted data governance.

3.What inconsistencies were detected in categorical values and how were they standardized?

Categorical inconsistencies typically involve variations in capitalization, spacing, or naming conventions within text-based dimensions like ship modes and regions. These textual discrepancies were resolved by applying standardization techniques such as custom aliases or string manipulation formulas like uppercase conversion and trimming. Standardizing these categorical fields ensured uniform grouping across all associated dimensions and metrics. Eliminating fragmented text entries prevented duplicate category classifications from skewing report outputs. As a result, dimensional filtering and categorical breakdowns operated with strict uniformity.

4.Describe how calculated fields helped in resolving data issues.

Calculated fields served as the primary programmatic mechanism for correcting data anomalies, handling null values, and generating custom evaluation flags directly within the environment. They enabled dynamic rule execution such as substituting missing text strings with default labels and evaluating numerical thresholds for outlier detection. By writing expressions like conditional statements, data stewards could automate quality control measures without altering the underlying raw source files. These expressions provided a flexible layer of data transformation that adapted seamlessly to changing reporting requirements. Overall, calculated fields streamlined the end-to-end data cleaning workflow by centralizing logic into reusable data elements.

5.How did your dashboard demonstrate the effectiveness of the data cleaning process?

The summary dashboard demonstrated the success of the data cleaning process by consolidating comparative visualizations, distribution histograms, and outlier scatter plots into a single unified interface. Side-by-side comparative views verified that valid entries remained uncorrupted while anomalies were successfully flagged and categorized. Metrics regarding record completeness and distribution patterns confirmed that data integrity standards were met across all transformed fields. Stakeholders could immediately review the impact of applied corrections through transparent graphical evidence and structured layout summaries. This comprehensive reporting interface validated that the refined dataset was fully prepared for advanced analysis.

RESULT:

The dataset was successfully cleaned and prepared for accurate analysis using Tableau and Power BI. Issues such as nulls, inconsistencies, and outliers were identified and resolved, improving data quality significantly.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		